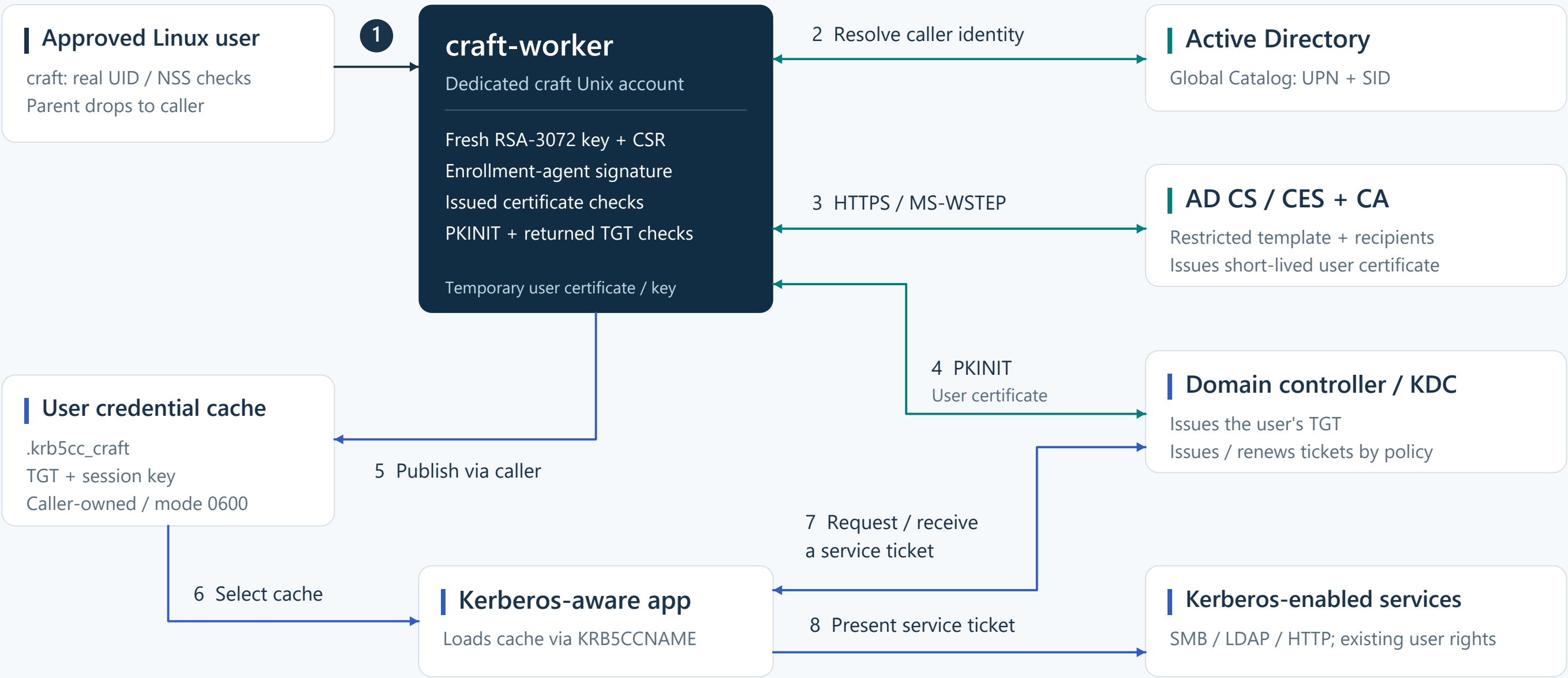


A temporary certificate. A reusable ticket workflow.

A ticket-granting ticket (TGT) lets approved Linux users request service tickets without storing AD passwords.

TRUSTED LINUX HOST

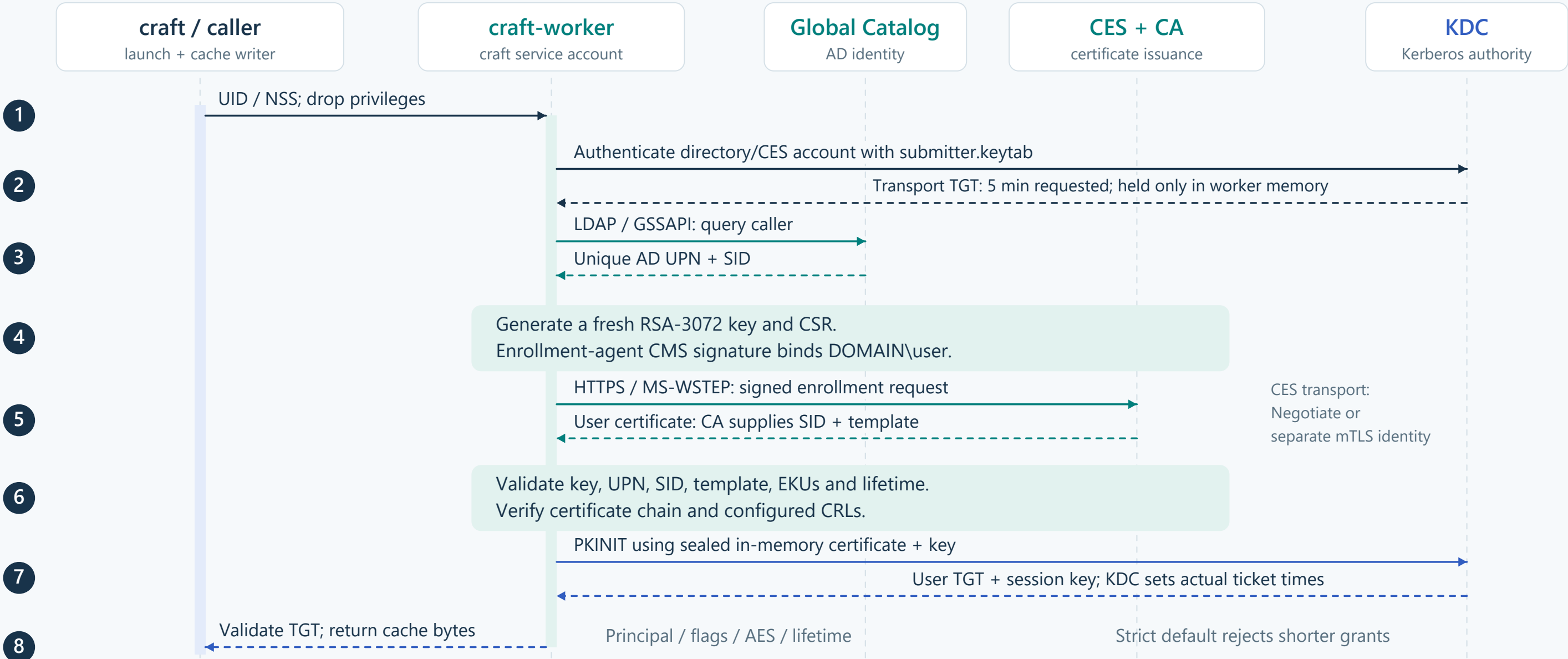
ACTIVE DIRECTORY + PKI



Trust comes from the managed host, enrollment credentials and CA restrictions. CRAFT performs no MFA challenge.

From a Linux account to a validated user TGT.

Successful exchange. Time flows downward; spacing represents order, not network latency.



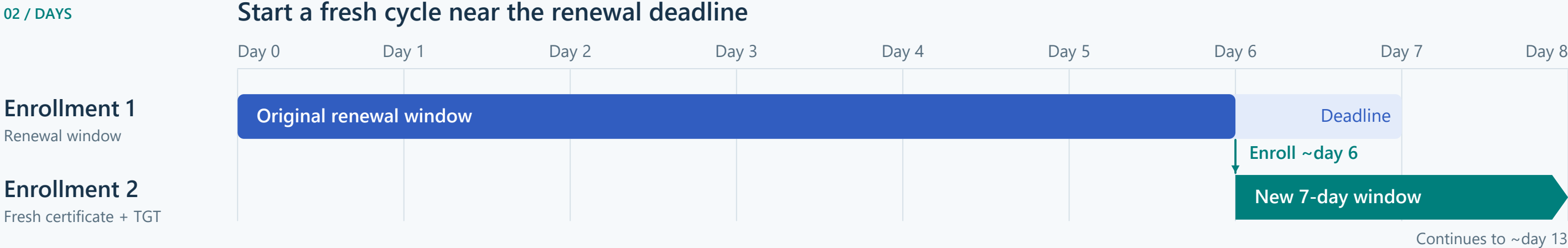
One acquisition, several independent clocks.

Default profile: request 10 h / 7 d. Certificate validity and KDC policy constrain the initial grant.

USER CERTIFICATE + PRIVATE KEY

Client residency lasts only for enrollment and PKINIT.

Certificate validity: ≤ 10 h total
CA backdating consumes this allowance.



Renewal uses the cached ticket and session key. Certificate expiry or cleanup does not revoke an issued ticket.

Keep a job authenticated, then stop with it.

craft-maintain runs as the caller, coordinates one shared cache per UID, and watches a live process.

START ONCE / WATCH A SHELL THAT STAYS ALIVE FOR THE JOB

```
job_pid=$$
cache=$(/usr/local/bin/craft-maintain --watch-pid "$job_pid") || exit "$?"
export KRB5CCNAME="$cache"
```

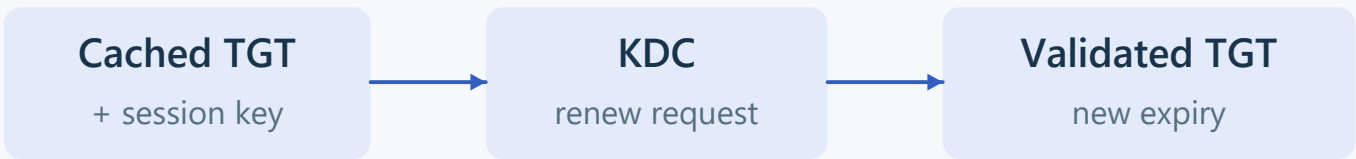
Returns when usable credentials and background maintenance are ready.

01 / RENEW

Use the current ticket

When a renewable TGT has $\leq 20\%$ of its lifetime left (capped at 2 h), and its expiry can extend.

A renewal error retries here; no immediate enrollment.



Renew-till never moves later.
Retain usable service tickets.

A due renewal takes priority during rollover, keeping usable credentials alive while enrollment is delayed.

02 / FRESH ENROLLMENT

Obtain fresh credentials

Missing or expired TGT; renewal deadline is near; or a ticket that cannot extend is nearing expiry.

Rollover margin: $1/7$ of the renewal window, capped at 1 day.



New renewal window.
Replace the shared cache.

Successful refresh

Publish atomically to .krb5cc_craft (caller-owned, mode 0600).

Refresh failure

Keep the existing cache and retry; the job continues.

RETRY

Renewal: ~30 s to 5 min + jitter.
Enrollment cooldown: 1 min to 1 h.

STOP

Watched PID exits, duration ends or signal.
Maintainer stops; the shared cache remains.

APPLICATION

Reload credentials when authenticating again.
Access follows the user's service permissions.